

Airport Flight Performance Analytics

1. Final Business Insights and Recommendations

We reviewed the final Power BI KPIs and analytical results to identify the major operational patterns in the airport operations dataset. The analysis focused on Total Flights, Total Passengers, Average Baggage Delay, Average Total Delay, Flight Delay Percentage, and On-Time Performance.

The KPI results were used to understand overall airport performance and identify areas that may require operational attention. We compared flight performance across Aircraft_Type and Status and examined delay-related measures to understand how operational delays affect overall performance.

Based on the analysis, the following business recommendations were prepared:

- **Monitor delay KPIs regularly:** Average Total Delay and Flight Delay Percentage should be monitored to identify changes in operational performance.
- **Focus on baggage operations:** Baggage delay should be tracked separately because it can contribute to overall turnaround and passenger-service issues.
- **Compare aircraft performance:** Aircraft_Type-wise analysis can help identify categories that show higher delays or operational variation.
- **Analyze flight status:** Comparing On-Time, Delayed, and other statuses helps management understand the overall flight performance pattern.
- **Use time-based analysis:** Flight_Date and Month trends can help identify periods with higher operational pressure and support better resource planning.
- **Use passenger volume for planning:** Passenger counts can support decisions related to gates, baggage handling, ground staff, and other airport resources.
- **Use the Power BI dashboard for monitoring:** The interactive dashboard can provide management with a quick view of KPIs and allow users to drill down into specific aircraft types, statuses, and time periods.

2. Dataset Limitations:

We also documented the limitations of the dataset before presenting the final conclusions. The dataset contains simulated/realistic airport operational information and should therefore be interpreted within the scope of the project rather than treated as a complete representation of all real-world airport operations.

The main limitations identified were:

- The analysis depends on the quality and completeness of the available dataset.
- Missing values and inconsistent records identified during profiling required cleaning and transformation.
- Some values were standardized during the cleaning process, so the cleaned dataset may differ from the original dataset.
- Outlier treatment can influence statistical measures and analytical results.
- The dataset represents a specific operational sample and may not represent every airport, airline, route, or time period.

- Correlation between variables does not necessarily mean that one variable directly causes another.
- The analysis is based on the available columns and does not include every possible factor that can affect flight delays.
- Business recommendations should therefore be considered as data-driven suggestions, not absolute operational decisions.
- The results should be revalidated with updated operational data before being used for real-world decision-making.